

Problem 1. Parametrized Paths. Consider the following parametrized path in $\mathbb{R}^2$:

$$\mathbf{r}(t) = (x(t), y(t)) = (2 + \cos t, 1 + \sin t).$$

- (a) Compute the velocity vector $\mathbf{r}'(t)$ and the speed $\|\mathbf{r}'(t)\|$ at time t .
- (b) Compute the arc length between times $t = 0$ and $t = \pi/2$.

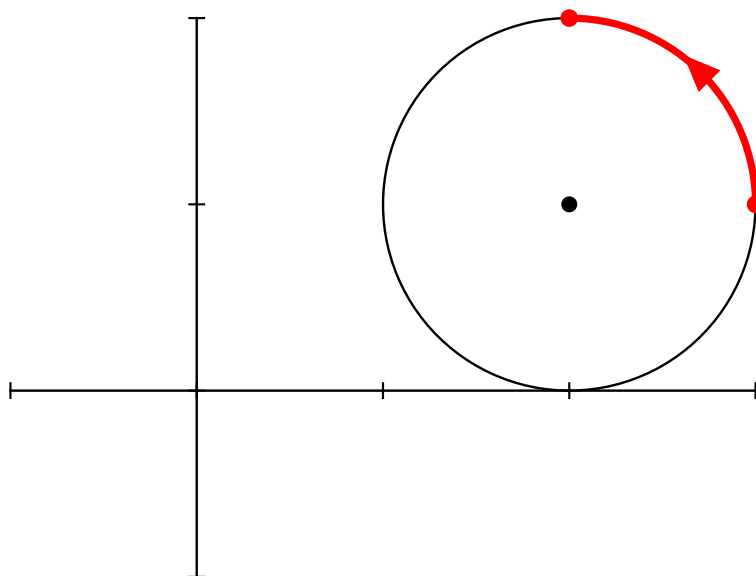
(a): The velocity vector is $\mathbf{r}'(t) = \langle 0 - \sin t, 0 + \cos t \rangle = \langle -\sin t, \cos t \rangle$ and the speed is

$$\|\mathbf{r}'(t)\| = \sqrt{(-\sin t)^2 + (\cos t)^2} = \sqrt{\sin^2 t + \cos^2 t} = \sqrt{1} = 1.$$

(b): Arc length is the integral of speed:

$$\text{arc length} = \int_0^{\pi/2} \|\mathbf{r}'(t)\| dt = \int_0^{\pi/2} 1 dt = t \Big|_0^{\pi/2} = \pi/2 - 0 = \pi/2.$$

Alternate solution. As t goes from 0 to 2π , the point $\mathbf{r}(t)$ travels counterclockwise around a circle of radius 1 centered at the point $(2, 1)$. The circumference of a unit circle is 2π . Between $t = 0$ and $t = \pi/2$ only one quarter of the circumference is traveled. Hence the arc length traveled is one quarter of the circumference, or $\pi/2$. Here is a picture:



Problem 2. Dot Product. Consider three vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in $\mathbb{R}^3$ satisfying

$$\mathbf{u} \bullet \mathbf{u} = 1, \quad \mathbf{v} \bullet \mathbf{v} = 4, \quad \mathbf{u} \bullet \mathbf{v} = 2.$$

- (a) Compute $\|\mathbf{u} - \mathbf{v}\|^2$.
- (b) Let θ be the angle between vectors $\mathbf{u}$ and $\mathbf{v}$, measured tail to tail. Compute $\cos \theta$.

(a): For any vector $\mathbf{x}$ we have $\|\mathbf{x}\|^2 = \mathbf{x} \bullet \mathbf{x}$. Hence

$$\|\mathbf{u} - \mathbf{v}\|^2 = (\mathbf{u} - \mathbf{v}) \bullet (\mathbf{u} - \mathbf{v}) = \mathbf{u} \bullet \mathbf{u} - 2\mathbf{u} \bullet \mathbf{v} + \mathbf{v} \bullet \mathbf{v} = 1 - 4 + 4 = 1.$$

(b): Recall the generalized Pythagorean theorem

$$\mathbf{u} \bullet \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta.$$

Hence we have

$$\cos \theta = \frac{\mathbf{u} \bullet \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{\mathbf{u} \bullet \mathbf{v}}{\sqrt{\mathbf{u} \bullet \mathbf{u}} \sqrt{\mathbf{v} \bullet \mathbf{v}}} = \frac{2}{\sqrt{1} \sqrt{4}} = 1.$$

Interpretation. This tells us that $\theta = 0$, so $\mathbf{u}$ and $\mathbf{v}$ are pointing in the same direction. Here is a picture of $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{u} - \mathbf{v}$. Note that $\mathbf{v}$ has length 2, while $\mathbf{u}$ and $\mathbf{u} - \mathbf{v}$ have length 1.

